# Chapter 1. Preliminaries to Complex Analysis

Complex numbers, holomorphic functions, power series, integration along curves

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## Overview

We review the algebraic and analytic properties of complex numbers and some topology of the complex plane. We then define *holomorphicity*, the complex analytic version of differentiability, which leads to the Cauchy–Riemann equations and to power series. Finally we define curves and integrals along them, and prove a first result in the direction of Cauchy's theorem: if $f$ has a primitive (a holomorphic $F$ with $F' = f$), then $\int_\gamma f(z)\,dz = 0$ for every closed curve $\gamma$.

# 1 Complex numbers and the complex plane

## 1.1 Basic properties

A complex number has the form $z = x+iy$ with $x, y$ real and $i^2 = -1$; $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$ are its **real part** and **imaginary part**. Real numbers are those with zero imaginary part; a number with zero real part is **purely imaginary**. The set of complex numbers is $\mathbb{C}$, identified with $\mathbb{R}^2$ by $z = x + iy \leftrightarrow (x, y)$; the $x$- and $y$-axes are the **real axis** and **imaginary axis**.

Treating all numbers as real and using $i^2 = -1$: if $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$,

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2), \qquad z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2).$$

These operations are commutative, associative and distributive. Addition is vector addition in $\mathbb{R}^2$; multiplication is a rotation composed with a dilation (see the polar form below); multiplication by $i$ is rotation by $\pi/2$.

The **absolute value** of $z = x+iy$ is $|z| = (x^2+y^2)^{1/2}$, the euclidean distance from 0 to $(x, y)$. The triangle inequality $|z + w| \le |z| + |w|$ holds, and also $|\mathrm{Re}(z)| \le |z|$, $|\mathrm{Im}(z)| \le |z|$ and

$$||z| - |w|| \le |z - w|,$$

which follows from $|z| \le |z - w| + |w|$ and $|w| \le |z - w| + |z|$.

The **complex conjugate** of $z = x + iy$ is $\overline{z} = x - iy$ (reflection across the real axis). $z$ is real iff $z = \overline{z}$, purely imaginary iff $z = -\overline{z}$, and

$$\mathrm{Re}(z) = \frac{z + \overline{z}}{2}, \qquad \mathrm{Im}(z) = \frac{z - \overline{z}}{2i}, \qquad |z|^2 = z\overline{z}, \qquad \frac{1}{z} = \frac{\overline{z}}{|z|^2}\ (z \neq 0).$$

Any non-zero $z$ has a **polar form** $z = re^{i\theta}$ with $r > 0$, $\theta \in \mathbb{R}$, where $e^{i\theta} = \cos\theta + i\sin\theta$. Since $|e^{i\theta}| = 1$, $r = |z|$; $\theta$ is the **argument** $\arg z$, defined up to a multiple of $2\pi$, the counterclockwise angle from the positive real axis to the half-line through $z$. If $z = re^{i\theta}$ and $w = se^{i\varphi}$ then $zw = rse^{i(\theta+\varphi)}$: multiplication is a homothety (a rotation composed with a dilation).

## 1.2 Convergence

A sequence $\{z_n\}$ **converges** to $w \in \mathbb{C}$ if $\lim_{n\to\infty}|z_n - w| = 0$; we write $w = \lim z_n$. Since $|\cdot|$ on $\mathbb{C}$ is the euclidean distance on $\mathbb{R}^2$, this is convergence of the points in the plane; equivalently, the real and imaginary parts of $z_n$ converge to those of $w$. $\{z_n\}$ is a **Cauchy sequence** if $|z_n - z_m| \to 0$ as $n, m \to \infty$, i.e. for every $\epsilon > 0$ there is $N$ with $|z_n - z_m| < \epsilon$ for $n, m > N$. $\mathbb{R}$ is complete (every Cauchy sequence of reals converges), and $\{z_n\}$ is Cauchy iff its real and imaginary parts are; hence:

**Theorem 1.1**

$\mathbb{C}$, the complex numbers, is complete.

## 1.3 Sets in the complex plane

For $z_0 \in \mathbb{C}$ and $r > 0$, the **open disc** is $D_r(z_0) = \{z \in \mathbb{C} : |z - z_0| < r\}$, the **closed disc** is $\overline{D}_r(z_0) = \{z : |z - z_0| \leq r\}$, and their boundary is the circle $C_r(z_0) = \{z : |z - z_0| = r\}$. The unit disc is denoted $\mathbb{D} = \{z : |z| < 1\}$.

Given $\Omega \subset \mathbb{C}$, $z_0$ is an **interior point** of $\Omega$ if $D_r(z_0) \subset \Omega$ for some $r > 0$; the **interior** of $\Omega$ consists of its interior points. $\Omega$ is **open** if every point of $\Omega$ is an interior point, and **closed** if $\Omega^c = \mathbb{C} - \Omega$ is open. $z$ is a **limit point** of $\Omega$ if there are $z_n \in \Omega$, $z_n \neq z$, with $z_n \to z$; a set is closed iff it contains all its limit points. The **closure** $\overline{\Omega}$ is $\Omega$ together with its limit points, and the **boundary** $\partial\Omega$ is the closure minus the interior. $\Omega$ is **bounded** if $|z| < M$ for $z \in \Omega$; then $\operatorname{diam}(\Omega) = \sup_{z,w\in\Omega}|z - w|$. $\Omega$ is **compact** if it is closed and bounded.

**Theorem 1.2**

The set $\Omega \subset \mathbb{C}$ is compact if and only if every sequence $\{z_n\} \subset \Omega$ has a subsequence that converges to a point in $\Omega$.

**Proof [Supplement].** If $\Omega$ is closed and bounded, a sequence in $\Omega$ has bounded real and imaginary parts; by Bolzano–Weierstrass (twice) a subsequence converges, and its limit lies in $\Omega$ since $\Omega$ is closed. Conversely, if $\Omega$ were unbounded, choose $z_n \in \Omega$ with $|z_n| > n$; no subsequence converges. If $\Omega$ were not closed, some limit point $z \notin \Omega$ is the limit of $z_n \in \Omega$, and every subsequence converges to $z \notin \Omega$. ∎

An **open covering** of $\Omega$ is a family of open sets $\{U_\alpha\}$ with $\Omega \subset \bigcup_\alpha U_\alpha$.

**Theorem 1.3**

A set $\Omega$ is compact if and only if every open covering of $\Omega$ has a finite subcovering.

**Proof [Supplement].** Suppose $\Omega$ is compact and $\{U_\alpha\}$ is an open covering with no finite subcovering. $\Omega$ lies in a square $Q_0$; subdivide it into four equal squares; the parts of $\Omega$ in at least one of them, $Q_1$, cannot be finitely covered. Repeating, we get nested squares $Q_n$ with $\operatorname{diam}(Q_n) \to 0$ such that $\Omega \cap Q_n$ is not finitely covered. Pick $z_n \in \Omega \cap Q_n$; $\{z_n\}$ is Cauchy, converges to some $w \in \Omega$ ($\Omega$ is closed), and $w \in U_\alpha$ for some $\alpha$. Since $U_\alpha$ is open and $\operatorname{diam}(Q_n) \to 0$, $Q_n \subset U_\alpha$ for large $n$, a contradiction. Conversely, if every open covering has a finite subcovering, the discs $D_n(0)$ cover $\Omega$, so $\Omega$ is bounded; and if $z \notin \Omega$ were a limit point, the sets $\{w : |w - z| > 1/n\}$ would cover $\Omega$ with no finite subcovering. ∎

**Proposition 1.4**

If $\Omega_1 \supset \Omega_2 \supset \cdots \supset \Omega_n \supset \cdots$ is a sequence of non-empty compact sets in $\mathbb{C}$ with the property that $\operatorname{diam}(\Omega_n) \to 0$ as $n \to \infty$, then there exists a unique point $w \in \mathbb{C}$ such that $w \in \Omega_n$ for all $n$.

**Proof.** Choose $z_n \in \Omega_n$. The condition $\operatorname{diam}(\Omega_n) \to 0$ says precisely that $\{z_n\}$ is Cauchy, so it converges to some $w$. Since each $\Omega_n$ is compact and contains $z_m$ for $m \geq n$, $w \in \Omega_n$ for all $n$. If $w' \neq w$ had the same property, then $|w - w'| > 0$ would violate $\operatorname{diam}(\Omega_n) \to 0$. ∎

An open set $\Omega$ is **connected** if it cannot be written as $\Omega = \Omega_1 \cup \Omega_2$ with $\Omega_1, \Omega_2$ disjoint non-empty open sets. A connected open set in $\mathbb{C}$ is a **region**. A closed set $F$ is connected if it cannot be written as $F = F_1 \cup F_2$ with $F_1, F_2$ disjoint non-empty closed sets. Equivalently, an open set $\Omega$ is connected iff any two points of $\Omega$ can be joined by a curve in $\Omega$.

# 2 Functions on the complex plane

## 2.1 Continuous functions

$f$ defined on $\Omega$ is **continuous** at $z_0 \in \Omega$ if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(z) - f(z_0)| < \epsilon$ whenever $z \in \Omega$, $|z - z_0| < \delta$; equivalently, $z_n \to z_0$ implies $f(z_n) \to f(z_0)$. $f$ is continuous on $\Omega$ if continuous at every point. Sums and products of continuous functions are continuous. A function of $z = x + iy$ is continuous iff it is continuous as a function of the two real variables $x, y$. If $f$ is continuous, so is $z \mapsto |f(z)|$. $f$ **attains a maximum** at $z_0$ if $|f(z)| \leq |f(z_0)|$ for all $z \in \Omega$ (reversed for a minimum).

**Theorem 2.1**

A continuous function on a compact set $\Omega$ is bounded and attains a maximum and minimum on $\Omega$.

(The proof is the same as for functions of a real variable.)

## 2.2 Holomorphic functions

**Definition (holomorphic function)**

Let $\Omega$ be open and $f$ complex-valued on $\Omega$. $f$ is **holomorphic at** $z_0 \in \Omega$ if the quotient

$$\frac{f(z_0 + h) - f(z_0)}{h} \tag{1}$$

converges to a limit as $h \to 0$ ($h \in \mathbb{C}$, $h \neq 0$, $z_0 + h \in \Omega$). The limit is the **derivative** $f'(z_0)$. $f$ is **holomorphic on** $\Omega$ if it is holomorphic at every point of $\Omega$; if $C$ is closed, $f$ is holomorphic on $C$ if it is holomorphic on some open set containing $C$; $f$ holomorphic on all of $\mathbb{C}$ is **entire**.

In the limit $h$ may approach 0 from any direction. The terms *regular* or *complex differentiable* are also used. Holomorphic functions turn out to be infinitely complex differentiable and *analytic* (they have power series expansions), in contrast with real-differentiable functions.

**Example 1**

$f(z) = z$ is holomorphic on any open set with $f'(z) = 1$; any polynomial $p(z) = a_0 + a_1 z + \cdots + a_n z^n$ is entire, with $p'(z) = a_1 + \cdots + n a_n z^{n-1}$ (by Proposition 2.2).

**Example 2**

$1/z$ is holomorphic on any open set not containing 0, with $f'(z) = -1/z^2$.

**Example 3**

$f(z) = \overline{z}$ is not holomorphic: $\dfrac{f(z_0 + h) - f(z_0)}{h} = \dfrac{\overline{h}}{h}$, which has no limit as $h \to 0$ (it is 1 for real $h$ and $-1$ for purely imaginary $h$).

Clearly $f$ is holomorphic at $z_0$ iff there is $a \in \mathbb{C}$ with

$$f(z_0 + h) - f(z_0) - ah = h\psi(h), \tag{2}$$

where $\psi$ is defined for small $h$ and $\lim_{h\to 0}\psi(h)=0$; then $a=f'(z_0)$. Hence $f$ is continuous wherever it is holomorphic. Arguing as for one real variable, using (2):

**Proposition 2.2**

If $f$ and $g$ are holomorphic in $\Omega$, then:
(i) $f+g$ is holomorphic in $\Omega$ and $(f+g)'=f'+g'$.
(ii) $fg$ is holomorphic in $\Omega$ and $(fg)'=f'g+fg'$.
(iii) If $g(z_0)\neq 0$, then $f/g$ is holomorphic at $z_0$ and $(f/g)'=\dfrac{f'g-fg'}{g^2}$. Moreover, if $f:\Omega\to U$ and $g:U\to\mathbb{C}$ are holomorphic, the chain rule holds: $(g\circ f)'(z)=g'(f(z))f'(z)$ for all $z\in\Omega$.

**Proof [Supplement](chain rule).** Write $f(z+h)-f(z)=f'(z)h+h\psi_1(h)$ and $g(w+k)-g(w)=g'(w)k+k\psi_2(k)$ with $\psi_j\to 0$, $w=f(z)$. With $k=f(z+h)-f(z)\to 0$, $g(f(z+h))-g(f(z))=g'(w)k+k\psi_2(k)=g'(w)f'(z)h+h[g'(w)\psi_1(h)+(f'(z)+\psi_1(h))\psi_2(k)]$, and the bracket tends to 0. (i)–(iii) are similar. ∎

**Complex-valued functions as mappings.** $\overline{z}$ corresponds to $F:(x,y)\mapsto(x,-y)$, which is linear, hence real-differentiable to all orders; so real differentiability does not imply holomorphy. To $f=u+iv$ associate $F(x,y)=(u(x,y),v(x,y))$. $F$ is **differentiable** at $P_0=(x_0,y_0)$ if there is a linear $J:\mathbb{R}^2\to\mathbb{R}^2$ with

$$\frac{|F(P_0+H)-F(P_0)-J(H)|}{|H|}\to 0 \quad \text{as } |H|\to 0,\ H\in\mathbb{R}^2; \tag{3}$$

then $J$ is the Jacobian matrix $J_F=\begin{pmatrix}\partial u/\partial x & \partial u/\partial y\\ \partial v/\partial x & \partial v/\partial y\end{pmatrix}$. If $f$ is holomorphic at $z_0$, taking $h=h_1$ real in (1) gives $f'(z_0)=\frac{\partial f}{\partial x}(z_0)$, and taking $h=ih_2$ purely imaginary gives $f'(z_0)=\frac{1}{i}\frac{\partial f}{\partial y}(z_0)$. Hence $\frac{\partial f}{\partial x}=\frac{1}{i}\frac{\partial f}{\partial y}$, and separating real and imaginary parts (using $1/i=-i$) gives the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y}\qquad\text{and}\qquad\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}.$$

Define the operators

$$\frac{\partial}{\partial z}=\frac{1}{2}\Big(\frac{\partial}{\partial x}+\frac{1}{i}\frac{\partial}{\partial y}\Big),\qquad \frac{\partial}{\partial \overline{z}}=\frac{1}{2}\Big(\frac{\partial}{\partial x}-\frac{1}{i}\frac{\partial}{\partial y}\Big).$$

**Proposition 2.3**

If $f$ is holomorphic at $z_0$, then

$$\frac{\partial f}{\partial \overline{z}}(z_0)=0\qquad\text{and}\qquad f'(z_0)=\frac{\partial f}{\partial z}(z_0)=2\frac{\partial u}{\partial z}(z_0).$$

Also, if we write $F(x,y)=f(z)$, then $F$ is differentiable in the sense of real variables, and $\det J_F(x_0,y_0)=|f'(z_0)|^2$.

**Proof.** Taking real and imaginary parts, the Cauchy–Riemann equations are equivalent to $\partial f/\partial\overline{z}=0$. By the observation above, $f'(z_0)=\frac{1}{2}(\frac{\partial f}{\partial x}(z_0)+\frac{1}{i}\frac{\partial f}{\partial y}(z_0))=\frac{\partial f}{\partial z}(z_0)$, and the Cauchy–Riemann equations give $\partial f/\partial z=2\partial u/\partial z$. For differentiability of $F$: if $H=(h_1,h_2)$ and $h=h_1+ih_2$, the Cauchy–Riemann equations imply

$$J_F(x_0,y_0)(H)=\Big(\frac{\partial u}{\partial x}-i\frac{\partial u}{\partial y}\Big)(h_1+ih_2)=f'(z_0)h,$$

identifying complex numbers with pairs of reals. Finally, by the Cauchy–Riemann equations,

$$\det J_F(x_0,y_0) = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial v}{\partial x}\frac{\partial u}{\partial y} = \Big(\frac{\partial u}{\partial x}\Big)^2 + \Big(\frac{\partial u}{\partial y}\Big)^2 = \Big|2\frac{\partial u}{\partial z}\Big|^2 = |f'(z_0)|^2. \tag{4}$$

■

The converse:

**Theorem 2.4**

Suppose $f = u + iv$ is a complex-valued function defined on an open set $\Omega$. If $u$ and $v$ are continuously differentiable and satisfy the Cauchy–Riemann equations on $\Omega$, then $f$ is holomorphic on $\Omega$ and $f'(z) = \partial f/\partial z$.

**Proof.** Write

$$u(x+h_1,y+h_2)-u(x,y) = \frac{\partial u}{\partial x}h_1+\frac{\partial u}{\partial y}h_2+|h|\psi_1(h), \qquad v(x+h_1,y+h_2)-v(x,y) = \frac{\partial v}{\partial x}h_1+\frac{\partial v}{\partial y}h_2+|h|\psi_2(h),$$

with $\psi_j(h) \to 0$ as $|h| \to 0$, $h = h_1 + ih_2$ (differentiability follows from continuity of the partials). Using the Cauchy–Riemann equations,

$$f(z+h) - f(z) = \Big(\frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}\Big)(h_1 + ih_2) + |h|\psi(h),$$

with $\psi = \psi_1 + i\psi_2 \to 0$. So $f$ is holomorphic and $f'(z) = 2\partial u/\partial z = \partial f/\partial z$. ■

## 2.3 Power series

The complex exponential is $e^z = \sum_{n=0}^{\infty} z^n/n!$. For real $z$ this is the usual exponential, and the series converges absolutely for every $z \in \mathbb{C}$, since $|z^n/n!| = |z|^n/n!$ and $\sum |z|^n/n! = e^{|z|} < \infty$; the estimate shows uniform convergence on every disc. We will see that $e^z$ is entire and its derivative is obtained termwise: $(e^z)' = \sum_{n=0}^{\infty} nz^{n-1}/n! = \sum_{m=0}^{\infty} z^m/m! = e^z$. The geometric series $\sum z^n$ converges absolutely only in $|z| < 1$, with sum $1/(1-z)$ (holomorphic on $\mathbb{C} - \{1\}$), since $\sum_{n=0}^{N} z^n = \frac{1-z^{N+1}}{1-z}$ and $z^{N+1} \to 0$ for $|z| < 1$.

A **power series** is an expansion $\sum_{n=0}^{\infty} a_n z^n$ with $a_n \in \mathbb{C}$ (5). If it converges absolutely at $z_0$, it converges for all $|z| \le |z_0|$.

**Theorem 2.5**

Given a power series $\sum_{n=0}^{\infty} a_n z^n$, there exists $0 \le R \le \infty$ such that:
(i) If $|z| < R$ the series converges absolutely.
(ii) If $|z| > R$ the series diverges. Moreover, if we use the convention that $1/0 = \infty$ and $1/\infty = 0$, then $R$ is given by Hadamard's formula

$$1/R = \limsup |a_n|^{1/n}.$$

$R$ is the **radius of convergence** and $|z| < R$ the **disc of convergence**; $R = \infty$ for $e^z$ and $R = 1$ for the geometric series.

**Proof.** Let $L = 1/R$ and suppose $L \neq 0, \infty$ (these cases are easy). If $|z| < R$, choose $\epsilon > 0$ with $(L+\epsilon)|z| = r < 1$. By definition of $L$, $|a_n|^{1/n} \le L + \epsilon$ for large $n$, so $|a_n||z|^n \le \{(L+\epsilon)|z|\}^n = r^n$, and comparison with $\sum r^n$ gives convergence. If $|z| > R$, a similar argument gives a sequence of terms whose absolute values tend to infinity, so the series diverges. ■

**Remark**

On the boundary $|z| = R$ either convergence or divergence can occur.

Further entire power series are

$$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}, \qquad \sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!},$$

agreeing with the usual functions for real $z$, and related to the exponential by the **Euler formulas**

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

**Theorem 2.6**

The power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ defines a holomorphic function in its disc of convergence. The derivative of $f$ is also a power series obtained by differentiating term by term the series for $f$, that is, $f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$. Moreover, $f'$ has the same radius of convergence as $f$.

**Proof.** Since $\lim n^{1/n} = 1$, $\limsup |a_n|^{1/n} = \limsup |n a_n|^{1/n}$, so $\sum a_n z^n$ and $\sum n a_n z^n$, hence $\sum n a_n z^{n-1}$, have the same radius of convergence (Hadamard's formula).

Let $g(z) = \sum n a_n z^{n-1}$, let $R$ be the radius of convergence, and $|z_0| < r < R$. Write $f(z) = S_N(z) + E_N(z)$ with $S_N(z) = \sum_{n=0}^{N} a_n z^n$, $E_N(z) = \sum_{n=N+1}^{\infty} a_n z^n$. If $|z_0 + h| < r$,

$$\frac{f(z_0+h) - f(z_0)}{h} - g(z_0) = \Big(\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\Big) + (S_N'(z_0) - g(z_0)) + \frac{E_N(z_0+h) - E_N(z_0)}{h}.$$

Since $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$,

$$\Big|\frac{E_N(z_0+h) - E_N(z_0)}{h}\Big| \le \sum_{n=N+1}^{\infty} |a_n| \Big|\frac{(z_0+h)^n - z_0^n}{h}\Big| \le \sum_{n=N+1}^{\infty} |a_n| n r^{n-1},$$

the tail of a convergent series (as $g$ converges absolutely on $|z| < R$). Given $\epsilon > 0$ choose $N_1$ so that $N > N_1$ makes this $< \epsilon$, and $N_2$ so that $N > N_2$ gives $|S_N'(z_0) - g(z_0)| < \epsilon$. Fix $N > \max(N_1, N_2)$; since the derivative of a polynomial is obtained termwise, there is $\delta > 0$ with $\big|\frac{S_N(z_0+h) - S_N(z_0)}{h} - S_N'(z_0)\big| < \epsilon$ for $|h| < \delta$. Then $\big|\frac{f(z_0+h) - f(z_0)}{h} - g(z_0)\big| < 3\epsilon$ for $|h| < \delta$. ∎

**Corollary 2.7**

A power series is infinitely complex differentiable in its disc of convergence, and the higher derivatives are also power series obtained by termwise differentiation.

A power series centered at $z_0$ is $f(z) = \sum a_n (z - z_0)^n$; with $g(w) = \sum a_n w^n$, $f(z) = g(z - z_0)$, so everything about $g$ transfers, e.g. $f'(z) = \sum n a_n (z - z_0)^{n-1}$.

**Definition (analytic function)**

$f$ defined on an open set $\Omega$ is **analytic** (has a power series expansion) at $z_0 \in \Omega$ if there is a power series $\sum a_n (z - z_0)^n$ with positive radius of convergence such that $f(z) = \sum a_n (z - z_0)^n$ for $z$ near $z_0$. If this holds at every point of $\Omega$, $f$ is analytic on $\Omega$.

By Theorem 2.6, analytic functions are holomorphic; the converse (Chapter 2) justifies using the two terms interchangeably.

# 3 Integration along curves

A **parametrized curve** is a function $z(t)$ from a closed interval $[a,b] \subset \mathbb{R}$ to $\mathbb{C}$. It is **smooth** if $z'(t)$ exists and is continuous on $[a,b]$ and $z'(t) \neq 0$; at $t = a, b$, $z'$ means the one-sided derivatives

$$z'(a) = \lim_{h\to 0,\, h>0} \frac{z(a+h)-z(a)}{h}, \qquad z'(b) = \lim_{h\to 0,\, h<0} \frac{z(b+h)-z(b)}{h}.$$

It is **piecewise-smooth** if $z$ is continuous on $[a,b]$ and smooth on intervals $[a_k, a_{k+1}]$ for some $a = a_0 < a_1 < \cdots < a_n = b$ (one-sided derivatives at the $a_k$ may differ). Two parametrizations $z : [a,b] \to \mathbb{C}$, $\tilde{z} : [c,d] \to \mathbb{C}$ are **equivalent** if there is a continuously differentiable bijection $s \mapsto t(s)$ of $[c,d]$ onto $[a,b]$ with $t'(s) > 0$ and $\tilde{z}(s) = z(t(s))$; $t' > 0$ means orientation is preserved. The equivalence class determines a smooth curve $\gamma \subset \mathbb{C}$, the image of $[a,b]$ with the orientation given by $z$. $\gamma^-$ is $\gamma$ with orientation reversed, parametrized e.g. by $z^-(t) = z(b+a-t)$. The **end-points** $z(a)$, $z(b)$ do not depend on the parametrization; $\gamma$ begins at $z(a)$ and ends at $z(b)$. $\gamma$ is **closed** if $z(a) = z(b)$, and **simple** if $z(t) \neq z(s)$ unless $s = t$ (for closed curves: unless $s = t$, or $s = a$ and $t = b$). For brevity, a *curve* means a piecewise-smooth curve.

The circle $C_r(z_0)$ has the **positive orientation** (counterclockwise) given by $z(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$, and the negative orientation given by $z(t) = z_0 + re^{-it}$. $C$ denotes a general positively oriented circle.

**Definition (integral along a curve; length)**

For a smooth curve $\gamma$ parametrized by $z : [a,b] \to \mathbb{C}$ and $f$ continuous on $\gamma$,

$$\int_\gamma f(z)\,dz = \int_a^b f(z(t))z'(t)\,dt.$$

For piecewise-smooth $\gamma$, $\int_\gamma f(z)\,dz = \sum_{k=0}^{n-1} \int_{a_k}^{a_{k+1}} f(z(t))z'(t)\,dt$. The **length** of a smooth curve is $\text{length}(\gamma) = \int_a^b |z'(t)|\,dt$, and that of a piecewise-smooth curve is the sum of the lengths of its smooth parts.

These are independent of the parametrization: if $\tilde{z}$ is equivalent to $z$, by change of variables and the chain rule

$$\int_a^b f(z(t))z'(t)\,dt = \int_c^d f(z(t(s)))z'(t(s))t'(s)\,ds = \int_c^d f(\tilde{z}(s))\tilde{z}'(s)\,ds.$$

**Proposition 3.1**

Integration of continuous functions over curves satisfies the following properties:
(i) It is linear: if $\alpha, \beta \in \mathbb{C}$, then $\int_\gamma (\alpha f(z) + \beta g(z))\,dz = \alpha \int_\gamma f(z)\,dz + \beta \int_\gamma g(z)\,dz$.
(ii) If $\gamma^-$ is $\gamma$ with the reverse orientation, then $\int_\gamma f(z)\,dz = -\int_{\gamma^-} f(z)\,dz$.
(iii) One has the inequality $\left|\int_\gamma f(z)\,dz\right| \leq \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$.

**Proof.** (i) follows from linearity of the Riemann integral.

(ii) [Supplement]: with $z^-(t) = z(a+b-t)$, $\int_a^b f(z^-(t))(z^-)'(t)\,dt = -\int_a^b f(z(a+b-t))z'(a+b-t)\,dt = -\int_a^b f(z(s))z'(s)\,ds$. (iii): $\left|\int_\gamma f(z)\,dz\right| \leq \sup_{t\in[a,b]}|f(z(t))| \int_a^b |z'(t)|\,dt \leq \sup_{z\in\gamma}|f(z)| \cdot \text{length}(\gamma)$. ■

A **primitive** for $f$ on $\Omega$ is a function $F$ holomorphic on $\Omega$ with $F'(z) = f(z)$ for all $z \in \Omega$.

**Theorem 3.2**

If a continuous function $f$ has a primitive $F$ in $\Omega$, and $\gamma$ is a curve in $\Omega$ that begins at $w_1$ and ends at $w_2$, then $\int_\gamma f(z)\,dz = F(w_2) - F(w_1)$.

**Proof.** If $\gamma$ is smooth with parametrization $z : [a, b] \to \mathbb{C}$, $z(a) = w_1$, $z(b) = w_2$, then by the chain rule and the fundamental theorem of calculus

$$\int_\gamma f(z)\,dz = \int_a^b F'(z(t))z'(t)\,dt = \int_a^b \frac{d}{dt}F(z(t))\,dt = F(z(b)) - F(z(a)).$$

If $\gamma$ is piecewise-smooth we get a telescoping sum: $\sum_{k=0}^{n-1}[F(z(a_{k+1})) - F(z(a_k))] = F(z(b)) - F(z(a))$. ■

**Corollary 3.3**

If $\gamma$ is a closed curve in an open set $\Omega$, and $f$ is continuous and has a primitive in $\Omega$, then $\oint_\gamma f(z)\,dz = 0$.

This is immediate since the end-points of a closed curve coincide. For example, $1/z$ has no primitive in $\mathbb{C} - \{0\}$: for the unit circle $C$, $z(t) = e^{it}$,

$$\oint_C f(z)\,dz = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}\,dt = 2\pi i \neq 0.$$

This innocent calculation lies at the heart of the theory.

**Corollary 3.4**

If $f$ is holomorphic in a region $\Omega$ and $f' = 0$, then $f$ is constant.

**Proof.** Fix $w_0 \in \Omega$; it suffices to show $f(w) = f(w_0)$ for all $w \in \Omega$. Since $\Omega$ is connected, there is a curve $\gamma$ joining $w_0$ to $w$. $f$ is a primitive of $f'$, so $\int_\gamma f'(z)\,dz = f(w) - f(w_0)$ by Theorem 3.2. But $f' = 0$, so the integral is 0 and $f(w) = f(w_0)$. ■